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Best effort service using a non-guaranteed bit rate (non-GBR) bearer

Abstract

Solutions for providing best effort service for voice calls (including video calls) using a non-guaranteed bit rate (GBR) bearer include: receiving, at a proxy node (e.g., a proxy-call session control function (P-CSCF)), a request to set up a voice session for a UE; requesting, by the proxy node, a policy node (e.g., a policy and charging rules function (PCRF) or a policy control function (PCF)) to set up a GBR data pipe (e.g., a dedicated bearer, a quality of service (QoS) class identifier (QCI) 1 bearer, or a fifth generation (5G) QoS identifier (5QI) 1 bearer); based at least on an error condition (e.g., a timeout or error message), instructing, by the proxy node, the UE to set up the voice session using an existing non-GBR data pipe (e.g., a default bearer, a QCI-5 bearer, or a 5QI-5 bearer).

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References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
2016/0212668	12/2015	Castro Castro	N/A	H04L 47/80
2020/0053135	12/2019	Liang	N/A	H04L 12/1407
2022/0386164	12/2021	Lee	N/A	H04W 28/0273
2023/0007557	12/2022	Zhu	N/A	H04W 36/00226
2023/0139372	12/2022	Sinha	370/259	H04L 65/1045
2023/0283645	12/2022	Atarius	370/328	H04L 69/165

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Background/Summary

BACKGROUND

(1) When a cellular device, such as a user equipment (UE), connects to a cellular network, such as a fourth generation (4G) or fifth generation (5G) cellular network, two data pipes are typically set up. One is a data default bearer and one is an internet protocol (IP) multimedia system (IMS) default bearer. The IMS default bearer is used for session initiation protocol (SIP) signaling, which is used for setting up voice calls.

(2) When the UE attempts to place a voice call, such as a voice over long-term evolution (VoLTE) call in 4G, a voice over new radio (VoNR) call in 5G, or a video call, SIP signaling over the IMS default bearer in the control plane is used to set up an IMS dedicated bearer for a call data pipe. If a new IMS dedicated bearer cannot be set up in time (e.g., 10 to 15 seconds) when a UE is attempting to place a new voice call, the call will be unsuccessful. If a new IMS dedicated bearer cannot be set up in time during a handover, the call will be dropped. Both scenarios degrade user experience.

SUMMARY

(3) The following summary is provided to illustrate examples disclosed herein, but is not meant to limit all examples to any particular configuration or sequence of operations.

(4) Solutions for providing best effort service for voice calls (including video calls) using a non-guaranteed bit rate (GBR) bearer include: receiving, at a proxy node (e.g., a proxy-call session control function (P-CSCF)), a request to set up a voice session for a UE; requesting, by the proxy node, a policy node (e.g., a policy and charging rules function (PCRF) or a policy control function (PCF)) to set up a GBR data pipe (e.g., a dedicated bearer, a quality of service (QoS) class

identifier (QCI) 1 bearer, or a fifth generation (5G) QoS identifier (5QI) 1 bearer); based at least on an error condition (e.g., a timeout or error message), instructing, by the proxy node, the UE to set up the voice session using an existing non-GBR data pipe (e.g., a default bearer, a QCI-5 bearer, or a 5QI-5 bearer).

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) The disclosed examples are described below with reference to the accompanying drawing figures listed below, wherein:
- (2) FIG. 1 illustrates an exemplary arrangement that advantageously provides best effort service for voice calls (including video calls) using a non-guaranteed bit rate (GBR) bearer;
- (3) FIG. 2 illustrates exemplary data pipes used in the arrangement of FIG. 1;
- (4) FIG. 3 illustrates a message sequence diagram showing exemplary messages that may be used in the arrangement of FIG. 1;
- (5) FIG. 4 illustrates a flowchart of exemplary operations associated with the arrangement of FIG. 1;
- (6) FIG. 5 illustrates another flowchart of exemplary operations associated with the arrangement of FIG. 1; and
- (7) FIG. 6 illustrates a block diagram of a computing device suitable for implementing various aspects of the disclosure.
- (8) Corresponding reference characters indicate corresponding parts throughout the drawings. References made throughout this disclosure, relating to specific examples, are provided for illustrative purposes, and are not meant to limit all implementations or to be interpreted as excluding the existence of additional implementations that also incorporate the recited features.

DETAILED DESCRIPTION

- (9) Solutions for providing best effort service for voice calls (including video calls) using a non-guaranteed bit rate (GBR) bearer include: receiving, at a proxy node (e.g., a proxy-call session control function (P-CSCF)), a request to set up a voice session for a UE; requesting, by the proxy node, a policy node (e.g., a policy and charging rules function (PCRF) or a policy control function (PCF)) to set up a GBR data pipe (e.g., a dedicated bearer, a quality of service (QoS) class identifier (QCI) 1 bearer, or a fifth generation (5G) QoS identifier (5QI) 1 bearer); based at least on an error condition (e.g., a timeout or error message), instructing, by the proxy node, the UE to set up the voice session using an existing non-GBR data pipe (e.g., a default bearer, a QCI-5 bearer, or a 5QI-5 bearer).
- (10) In some examples, the voice session is a voice over long-term evolution (VoLTE) call in a fourth generation (4G) cellular network, a voice over new radio (VoNR) call in a 5G cellular network, another wireless voice call type (e.g., in a different cellular generation, or a video call (included within “voice call” herein)). In some examples, the GBR data pipe is an internet protocol (IP) multimedia system (IMS) dedicated bearer, and the non-GBR data pipe is an IMS default bearer.
- (11) Aspects of the disclosure improve the reliability of cellular communications by permitting voice calls to occur over a non-GBR default bearer when the wireless network is unable to set up a GBR dedicated bearer (e.g., based at least on an error condition, instructing, by the proxy node, the UE to set up the voice session using an existing non-GBR data pipe). This operation provides advantages that improve user experience. Whereas, previously, the voice call would not go through (i.e., the call would fail) or would drop during handover, aspects of the disclosure enable a voice call to go through or survive a handover, although with a channel condition (i.e., no GBR) that theoretically may result in lower call quality. In experiments, however, the difference in call quality

was generally unnoticeable by users.

(12) With reference now to the figures, FIG. 1 illustrates an arrangement **100** that advantageously provides best effort service for voice calls (including video calls) using a non-GBR bearer.

Arrangement **100** permits a UE **102** to transmit voice data packets and video call data packets for a voice session **106** in a call with a UE **104** using non-GBR data pipe **204** (e.g., a default bearer used as a signaling data pipe, see FIG. 2) when a requested GBR data pipe **206** (a dedicated bearer, which is typically used as a voice data pipe) cannot be set up in time to prevent abandonment or dropping of the call. In some examples, voice data packets and video call data packets use real-time transport protocol (RTP), meaning that RTP packets will be sent over non-GBR data pipe **204** if GBR data pipe **206** cannot be set up in time.

(13) UE **102** communicates with a serving base station **112** of a (cellular) wireless network **110** over an air interface **114**. Wireless network **110** includes an access node **122**, a session management node **124**, a policy node **126**, a diameter routing agent (DRA) **128**, a packet routing node **132**, and a proxy node **134**. DRA **128** may be a DRA in 4G, although in 5G, DRA **128** may instead be a Service Communication Proxy (SCP) or be eliminated and replaced with a direct interface between packet routing node **132** and proxy node **134**. In some examples, proxy node **134** comprises a P-CSCF.

(14) For 4G, access node **122** may comprise a mobility management entity (MME), session management node **124** may comprise a packet data network gateway (PGW) control plane function (PGW-C) and a serving gateway (SGW) control plane function (SGW-C), policy node **126** may comprise a policy and charging rules function (PCRF), and packet routing node **132** may comprise a PGW user plane function (PGW-U) and an SGW user plane function (SGW-U). For 5G, access node **122** may comprise an access and mobility management function (AMF), session management node **124** may comprise a session management function (SMF), policy node **126** may comprise a policy control function (PCF), DRA **128** may be an SCP, and packet routing node **132** may comprise a user plane function (UPF).

(15) A cellular wireless network may employ multiple policy nodes (e.g., PCFs or PCRFs) that are grouped into destination realms (also known as diameter realms), each of which may include multiple policy nodes and a DRA/SCP. A DRA (e.g., DRA **128**) provides real-time routing capabilities to ensure that network messages are routed to the correct destinations. An SCP may employ a service based interface such as the N5 interface or a network application function (Naf) in a service based architecture (SBA).

(16) For voice and video calls, the route for the data packets passes from UE **104** over air interface **114**, through base station **112**, packet routing node **132**, proxy node **134**, and an IMS **150**, which includes an IMS access gateway (IMS-AGW **152**) to UE **104**. Signaling to set up voice session **106** passes from UE **104** over air interface **114**, through base station **112**, and access node **122**, to session management node **124**. Session management node **124** is communicatively coupled to packet routing node **132** and proxy node **134**, permitting session management node **124** to instruct packet routing node **132** and proxy node **134** to assign and release resources for voice session **106**.

(17) Turning briefly to FIG. 2, exemplary data pipes **202**, **204**, and **206**, used in arrangement **100**, are shown. When UE **102** connects to wireless network **110**, general data pipe **202** is set up as a data default bearer, as a QCI-9 bearer in 4G or a 5QI-9 bearer in 5G. General data pipe **202** is used for generic data (e.g., streaming video) and is a non-guaranteed bit rate (non-GBR) bearer with an allowance for a 100 millisecond (ms) packet delay and a 1/1,000,000 packet loss rate. Also, after UE **102** connects to wireless network **110**, a signaling data pipe, shown as non-GBR data pipe **204** is set up as an IMS default bearer (e.g., as a QCI-5 bearer in 4G or a 5QI-5 bearer in 5G). Non-GBR data pipe **204** is used for IMS signaling, such as session initiation protocol (SIP) signaling that is used for setting up voice session **106**, with an allowance for a 100 ms packet delay and a 1/1,000,000 packet loss rate. However, whereas packets in generic data pipe **202** have a priority of 9, packets in non-GBR data pipe **204** have a priority of 1 (highest priority). Typically, default

bearers (e.g., general data pipe **202** and non-GBR data pipe **204**) remain active while UE **102** remains attached to wireless network **110**, and each has its own IP address.

(18) In typical voice call scenarios, when UE **102** sends SIP signaling to wireless network **110**, a voice call data pipe (e.g., GBR data pipe **206**) is set up to transmit voice packets of voice session **106**, and released when voice session **106** ends (or is replaced by another GBR data pipe **206** during a handover). GBR data pipe **206** is set up as an IMS dedicated bearer, as a QCI-1 bearer in 4G or a 5QI-1 bearer in 5G for voice only. GBR data pipe **206** has an allowance for a 100 ms packet delay and a 1/100 packet loss rate with a priority of 2 (second highest priority). A dedicated bearer is linked to a default bearer to provide a dedicated tunnel for specific traffic (e.g., voice traffic) and shares the same IP address. For video calls, two IMS dedicated bearers are used, one for the voice data and one for the video data. Thus for video calls, GBR data pipe **206** represents both a QCI-1 bearer in 4G or a 5QI-1 bearer for the voice and also a QCI-2 bearer in 4G or a 5QI-2 bearer in 5G for the video.

(19) Returning to FIG. **1**, when UE **102** sends a request to initiate voice session **106** to wireless network **110** (e.g., to session management node **124**), it is expected that setup of GBR data pipe **206** may take time or may require multiple attempts. Thus, proxy node **134** has a timer **144** that allows for a timeout period prior to determining that an error condition exists and GBR data pipe **206** will not be set up in time.

(20) When proxy node **134** receives the request from UE **102**, it can wait until a SIP **183** response arrives for the session to be initiated, as proxy node **134** is able to set up a dedicated bearer over the initial request or at the response to the request (over SIP). Proxy node **134** also has a fail-over logic **142** that determines when to abandon setting up GBR data pipe **206** and instead instruct UE **102** to use non-GBR data pipe **204** for voice session **106**. In some examples, policy node **126** also has a policy **148** that permits using non-GBR data pipe **204** for voice session **106**. The set up and exchange of information for IMS **150** takes place over the IMS default bearer (e.g., non-GBR data pipe **204**).

(21) Different triggers exist for fail-over logic **142** to determine that an error condition exists. One trigger may be a timeout with timer **144**, prior to receiving a message that GBR data pipe **206** has been successfully set up. Another trigger is receiving an error message, for example from DRA **128**, indicating that GBR data pipe **206** will not be set up. In some examples, DRA **128** has its own timer **146**. If, while routing messages among the various nodes of wireless network **110**, DRA **128** does not receive a response indicating that GBR data pipe **206** has been or will be successfully set up, prior to a timeout with timer **146**, DRA **128** may send an error code to proxy node **134**. There may be multiple different failures in wireless network that prevent GBR data pipe **206** from being successfully set up within the allotted time. These include a failure at DRA **128**, a failure at policy node **126**, or a failure of policy node **126** to establish resources (e.g., with packet routing node **132**) for setting up GBR data pipe **206**.

(22) FIG. **3** illustrates a message sequence diagram **300** showing exemplary messages that may be used in arrangement **100**. UE **104** transmits a SIP invite **302** over non-GBR data pipe **204** (which is used for signaling), which is received by proxy node **134**. UE **102** does not request dedicated bearer setup, however, proxy node **134** identifies that the request for a voice session (or video call session) triggers the setup of a dedicated bearer. UE **102** waits for the setup. Proxy node **134** starts its timer **144** at timer mark **304**, and transmits a message **306** to DRA **128** that contains the subscriber ID for UE **102**, an application function (AF) identification (ID), an IMS application layer charging ID, and media information (including the media type and media stream description). In some examples, message **306** is an AAR message to a DRA over an Rx interface. In some examples, message **306** is an Npcf_Policy_Authorization_Create_Request for SCP over an N5 interface. DRA **128** (which may be an SCP in 5G) forwards this to policy node **126** as message **310** and starts its own timer **146** at timer mark **308**. In some examples, message **310** is an AAR message or an Npcf_Policy_Authorization message.

(23) Two different error condition scenarios are shown. In the first illustrated scenario, there is a network event **312** that prevents policy node **126** from allocating resources with packet routing node **132** (which would have used message **314**) to set up GBR data pipe **206**. As a result, there is a timeout **316** (e.g., a timer lapse of timer **146**), and DRA **128** transmits an error message **318** to proxy node **134**. In the second illustrated scenario, there is a network event **320** that prevents DRA **128** from routing messages necessary to set up GBR data pipe **206** (e.g., AAR message **310** or a confirmation that resources have been allocated by packet routing node **132**). As a result, there is a timeout **322** (e.g., a timer lapse of timer **144**). Other failure/error scenarios also exist.

(24) In either of the illustrated scenarios (or another failure/error scenario), fail-over logic **142** to determine that an error condition exists and instructs UE **102** with a message **324** to use non-GBR data pipe **204** for voice session **106**. Until this point, UE **102** had been waiting for the setup of GBR data pipe **206**, so message **324** terminates the waiting period for UE **102**. UE **102** then uses non-GBR data pipe **204** to send data packets for voice session **106** to IMS **150** (e.g., using RTP), which are then forwarded to UE **104**.

(25) FIG. 4 illustrates a flowchart **400** of exemplary operations associated with providing a voice call for UE **102** using voice session **106** over wireless network **110**. In some examples, at least a portion of flowchart **400** may be performed using one or more computing devices **600** of FIG. 6 (e.g., UE **102**, packet routing node **132**, proxy node **134**, DRA **128**, and policy node **126** may be examples of computing device **600**). In some examples, the wireless network comprises a 4G cellular network. In some examples, the wireless network comprises a 5G cellular network. In some examples, the voice session is selected from the list consisting of: a VoLTE call, a VoNR call, another wireless voice call type, and a video call.

(26) Flowchart **400** commences with UE **102** requesting voice session **106** (e.g., with SIP invite **302**) in operation **402**. Operation **404** includes receiving, at proxy node **134**, the request to set up voice session **106** (for UE **102**) with GBR data pipe **206**. This request may be received over the IMS non-GBR default bearer (e.g., non-GBR data pipe **204**), because voice signaling is routed on the default bearer (even if a dedicated bearer is later set up successfully).

(27) In some examples, proxy node **134** comprises a P-CSCF. In some examples, operation **404** includes proxy node **134** receiving the request from UE **102**, possibly via session management node **124**. In some examples, the request to set up voice session **106** comprises a mobile origination (MO) message, such as when UE **102** initiates a call to UE **104**. In some examples, the request to set up voice session **106** comprises a mobile termination (MT) message; such as when UE **102** receives a call from UE **104**. In some examples, the request to set up voice session **106** comprises a request for a new call. In some examples, the request to set up voice session **106** comprises at least a portion of a handover message. In some examples, the request to set up voice session **106** comprises a SIP message.

(28) Operation **406** includes requesting, by proxy node **134**, policy node **126** to set up GBR data pipe **206** and then proxy node **134** starts timer **144** at operation **408**. Operation **406** may include proxy node **134** requesting that policy node **126** allocate resources, and routing the request through DRA **128**. In some examples, the policy node comprises a PCRF. In some examples, the policy node comprises a PCF. In some examples, GBR data pipe **206** comprises a dedicated bearer. In some examples, GBR data pipe **206** comprises an IMS dedicated bearer. In some examples, GBR data pipe **206** comprises a QCI-1 bearer (plus a QCI-2 bearer for the video data, for video calls). In some examples, GBR data pipe **206** comprises a 5QI-1 or 5QI-2 bearer.

(29) DRA **128** receives and forwards the request (from proxy node **134** to policy node **126**) in operation **410** (e.g. as AAR messages **306** and **310**), and starts timer **146** at operation **412**. Decision operation **414** determines whether GBR data pipe **206** may be set up (e.g., resources are available and allocated). If so, proxy node **134** instructs UE **102** to set up and use GBR data pipe **206** for voice session **106** in operation **416**.

(30) While proxy node **134** is awaiting a message that GBR data pipe **206** may be set up, flowchart

400 moves to operation **418**, which includes attempting, by policy node **126**, to establish a resource (or resources) for voice session **106**. This may include, for example, attempting, by policy node **126**, to establish a resource for voice session **106** with packet routing node **132** (e.g., with message **314**). In some examples, packet routing node **132** comprises a PGW. In some examples, packet routing node **132** comprises a UPF.

(31) Decision operation **420** determines whether an error condition exists that warrants using GBR data pipe **206** for voice session **106**. In some examples, the error condition comprises a timeout prior to allocating a needed resource for GBR data pipe **206** (e.g., allocating resources on packet routing node **132**). Decision operation **420** remains looping while timer **144** remains running, or until an error message is received by proxy node **134** (e.g., error message **318** from DRA **128**). If proxy node **134** detects that timer **144** has lapsed (e.g., a timeout has occurred), this is one of the possible triggers for determining that an error condition has occurred and flowchart **400** moves to operation **428** (described below). Another trigger is an error message. In some examples, the error condition comprises an error condition selected from the list consisting of: a timeout (e.g., lapse of timer **144** or **146**) prior to allocating a needed resource for GBR data pipe **206**, an error message received by proxy node **134**, or a detected network condition as described below.

(32) Decision operation **422** is performed by another node in wireless network **110**, for example by DRA **128**. If resources for GBR data pipe **206** are not allocated prior to DRA **128** detecting a lapse of timer **146**, or DRA **128** detects another error condition, DRA **128** transmits an error message (e.g., error message **318**) to proxy node **134** in operation **424**. In some examples, the error condition comprises a timeout of timer **146** at DRA **128**. In some examples, the error condition comprises a network error rate exceeding an error threshold, such as a high bit error rate (BER), dropped packet rate, or dropped call rate. In some examples, the error condition comprises a network error performance metric failing to meet a performance threshold, such as throughput. Error conditions based on performance (e.g., a detected network condition) may be detected at other nodes within wireless network **110**, such as session management node **124**, or even proxy node **134** itself.

(33) If DRA **128** detects an error condition and transmits an error message to proxy node **134**, operation **424** also includes receiving, by proxy node **134**, the error message. Otherwise, operation **426** includes transmitting, by DRA **128**, a success message to proxy node **134**. In the event of successful allocation of resources for GBR data pipe **206**, flowchart **400** returns to decision operation **414** and then moves to operation **416**, described above.

(34) Returning to the error condition scenario of decision operation **422**. After operation **424** transmits the error message, flowchart **400** returns to decision operation **420**, which directs flowchart **400** into operation **428**. Operation **428** includes, based at least on an error condition, instructing, by proxy node **134**, UE **102** to set up voice session **106** using (existing) non-GBR data pipe **204**. In some examples, the non-GBR data pipe **204** comprises a default bearer. In some examples, the non-GBR data pipe **204** comprises an IMS default bearer. In some examples, the non-GBR data pipe **204** comprises a QCI-5 bearer. In some examples, the non-GBR data pipe **204** comprises a 5QI-5 bearer.

(35) Operation **430** includes (based at least on the instruction from proxy node **134**) setting up, by UE **102**, voice session **106** using the non-GBR data pipe **204**. Operation **432** includes UE **102** transmitting voice packets for voice session **106** over non-GBR data pipe **204**.

(36) FIG. 5 illustrates a flowchart **500** of exemplary operations associated with examples of arrangement **100**. In some examples, at least a portion of flowchart **500** may be performed using one or more computing devices **600** of FIG. 6. Flowchart **500** commences with operation **502**, which includes receiving, at a proxy node, a request to set up a voice session for a UE. Operation **504** includes requesting, by the proxy node, a policy node to set up a GBR data pipe. Operation **506** includes, based at least on an error condition, instructing, by the proxy node, the UE to continue setting up the voice session using an existing non-GBR data pipe.

(37) FIG. 6 illustrates a block diagram of computing device 600 that may be used as any component described herein that may require computational or storage capacity. Computing device 600 has at least a processor 602 and a memory 604 that holds program code 610, data area 620, and other logic and storage 630. Memory 604 is any device allowing information, such as computer executable instructions and/or other data, to be stored and retrieved. For example, memory 604 may include one or more random access memory (RAM) modules, flash memory modules, hard disks, solid-state disks, persistent memory devices, and/or optical disks. Program code 610 comprises computer executable instructions and computer executable components including any instructions necessary to perform operations described herein. Data area 620 holds any data necessary to perform operations described herein. Memory 604 also includes other logic and storage 630 that performs or facilitates other functions disclosed herein or otherwise required of computing device 600. An input/output (I/O) component 640 facilitates receiving input from users and other devices and generating displays for users and outputs for other devices. A network interface 650 permits communication over a network 660 with a remote node 670, which may represent another implementation of computing device 600. For example, a remote node 670 may represent another of the above-noted nodes within arrangement 100.

ADDITIONAL EXAMPLES

(38) An example method of providing a voice session over a wireless network comprises: receiving, at a proxy node, a request to set up a voice session for a UE; requesting, by the proxy node, a policy node to set up a GBR data pipe; based at least on an error condition, instructing, by the proxy node, the UE to set up the voice session using an existing non-GBR data pipe.

(39) An example system for providing a voice session over a wireless network, the system comprises: a processor; and a computer-readable medium storing instructions that are operative upon execution by the processor to: receive, at a proxy node, a request to set up a voice session for a UE; request, by the proxy node, a policy node to set up a GBR data pipe; based at least on an error condition, instruct, by the proxy node, the UE to set up the voice session using an existing non-GBR data pipe.

(40) One or more example computer storage devices has computer-executable instructions stored thereon, which, upon execution by a computer, cause the computer to perform operations comprising: receiving, at a proxy node, a request to set up a voice session for a UE; requesting, by the proxy node, a policy node to set up a GBR data pipe; based at least on an error condition, instructing, by the proxy node, the UE to set up the voice session using an existing non-GBR data pipe.

(41) Alternatively, or in addition to the other examples described herein, examples include any combination of the following: the voice session is selected from the list consisting of: a VoLTE call, a VoNR call, another wireless voice call type, and a video call; the wireless network comprises a 4G cellular network; the wireless network comprises a 5G cellular network; receiving, from the UE, the request to set up the voice session; the proxy node comprises a P-CSCF; the request to set up the voice session comprises an MO message; the request to set up the voice session comprises an MT message; the request to set up the voice session comprises a request for a new call; the request to set up the voice session comprises at least a portion of a handover message; the request to set up the voice session comprises a SIP message the policy node comprises a PCRF; the policy node comprises a PCF; the GBR data pipe comprises a dedicated bearer; the GBR data pipe comprises an IMS dedicated bearer; the GBR data pipe comprises a QCI-1 bearer; the GBR data pipe comprises a QCI-2 bearer; the GBR data pipe comprises a 5QI-1 bearer; the GBR data pipe comprises a 5QI-2 bearer; attempting, by the policy node, to establish a resource for the voice session; attempting, by the policy node, to establish a resource for the voice session with a packet routing node; the packet routing node comprises a PGW; the packet routing node comprises a UPF; detecting a lapse of a timer by the proxy node; detecting a lapse of a timer by the DRA; transmitting, by the DRA, an error message to the proxy node; receiving, by the proxy node, the

error message; the non-GBR data pipe comprises a default bearer; the non-GBR data pipe comprises an IMS default bearer; the non-GBR data pipe comprises a QCI-5 bearer; the error condition comprises a timeout prior to allocating a resource for the GBR data pipe; the error condition comprises a timeout of a timer at the proxy node; the error condition comprises a timeout of a timer at the DRA; the error condition comprises an error message received by the proxy node; the error condition comprises a network error rate exceeding an error threshold; the error condition comprises a network error performance metric failing to meet a performance threshold; setting up the voice session using the non-GBR data pipe; and based at least on the instruction from the proxy node, setting up the voice session using the non-GBR data pipe.

(42) The order of execution or performance of the operations in examples of the disclosure illustrated and described herein is not essential, unless otherwise specified. That is, the operations may be performed in any order, unless otherwise specified, and examples of the disclosure may include additional or fewer operations than those disclosed herein. For example, it is contemplated that executing or performing a particular operation before, contemporaneously with, or after another operation is within the scope of aspects of the disclosure. It will be understood that the benefits and advantages described above may relate to one embodiment or may relate to several embodiments. When introducing elements of aspects of the disclosure or the examples thereof, the articles “a,” “an,” “the,” and “said” are intended to mean that there are one or more of the elements. The terms “comprising,” “including,” and “having” are intended to be inclusive and mean that there may be additional elements other than the listed elements. The term “exemplary” is intended to mean “an example of.”

(43) Having described aspects of the disclosure in detail, it will be apparent that modifications and variations are possible without departing from the scope of aspects of the disclosure as defined in the appended claims. As various changes may be made in the above constructions, products, and methods without departing from the scope of aspects of the disclosure, it is intended that all matter contained in the above description and shown in the accompanying drawings shall be interpreted as illustrative and not in a limiting sense.

Claims

1. A method of providing a voice session over a wireless network, the method comprising: receiving, at a proxy-call session control function (P-CSCF) node, a request to set up a voice session for a user equipment (UE); transmitting, by the P-CSCF node to a diameter routing agent (DRA) or Service Communication Proxy (SCP), a request to set up a fifth generation (5G) quality of service (QoS) identifier 1 (5QI-1) or 5G QoS identifier 2 (5QI-2) bearer for the voice session of the UE, the request including an identifier (ID) assigned to the UE, wherein the 5QI-1 or 5QI-2 bearer is a guaranteed bit rate (GBR) data pipe of a 5G network, and wherein the P-CSCF node starts a timer upon transmitting the request; receiving, by the P-CSCF node, an indication of an error condition from the DRA or SCP prior to timeout of the timer, wherein the error condition corresponds to a network error rate exceeding an error threshold or a network performance metric failing to meet a performance threshold; and based at least on the error condition, sending, by the P-CSCF node, an instruction to the UE that instructs the UE to use an existing 5G QoS identifier 5 (5QI-5) bearer for the voice session of the UE, wherein the existing 5QI-5 bearer is a non-GBR data pipe of the 5G network.
2. The method of claim 1, wherein the 5QI-1 or 5QI-2 bearer comprises a dedicated bearer, and wherein the existing 5QI-5 bearer comprises a default bearer.
3. The method of claim 1, wherein the DRA or SCP forwards the request to a policy and charging rules function (PCRF).
4. The method of claim 1, wherein the DRA or SCP forwards the request to a policy control function (PCF).

5. The method of claim 1, wherein the voice session is a voice over long-term evolution (VoLTE) call.
6. The method of claim 1, wherein the voice session is a voice over new radio (VoNR) call.
7. The method of claim 1, wherein the voice session is a video call.
8. A system for providing a voice session over a wireless network, the system comprising: a processor; and a computer-readable medium storing instructions that are operative upon execution by the processor to: receive, at a proxy-call session control function (P-CSCF) node, a request to set up a voice session for a user equipment (UE); transmit, by the P-CSCF node to a diameter routing agent (DRA) or Service Communication Proxy (SCP), a request to set up a fifth generation (5G) quality of service (QoS) identifier 1 (5QI-1) or 5G QoS identifier 2 (5QI-2) bearer for the voice session of the UE, the request including an identifier (ID) assigned to the UE, wherein the 5QI-1 or 5QI-2 bearer is a guaranteed bit rate (GBR) data pipe of a 5G network, and wherein the P-CSCF node starts a timer upon transmitting the request; receive, by the P-CSCF node, an indication of an error condition from the DRA or SCP prior to timeout of the timer, wherein the error condition corresponds to a network error rate exceeding an error threshold or a network performance metric failing to meet a performance threshold; and based at least on the error condition, send, by the P-CSCF node, an instruction to the UE that instructs the UE to use an existing 5G QoS identifier 5 (5QI-5) bearer for the voice session of the UE, wherein the existing 5QI-5 bearer is a non-GBR data pipe of the 5G network.
9. The system of claim 8, wherein the 5QI-1 or 5QI-2 bearer comprises a dedicated bearer, and wherein the existing 5QI-5 bearer comprises a default bearer.
10. The system of claim 8, wherein the DRA or SCP forwards the request to a policy and charging rules function (PCRF).
11. The system of claim 8, wherein the DRA or SCP forwards the request to a policy control function (PCF).
12. The system of claim 8 wherein the voice session is a voice over long-term evolution (VoLTE) call or a voice over new radio (VoNR) call.
13. The system of claim 8, wherein the voice session is a video call.
14. One or more computer storage devices having computer-executable instructions stored thereon, which, upon execution by a computer, cause the computer to perform operations comprising: receiving, at a proxy-call session control function (P-CSCF) node, a request to set up a voice session for a user equipment (UE); transmitting, by the P-CSCF node to a diameter routing agent (DRA) or Service Communication Proxy (SCP), a request to set up a fifth generation (5G) quality of service (QoS) identifier 1 (5QI-1) or 5G QoS identifier 2 (5QI-2) bearer for the voice session of the UE, the request including an identifier (ID) assigned to the UE, wherein the 5QI-1 or 5QI-2 bearer is a guaranteed bit rate (GBR) data pipe of a 5G network, and wherein the P-CSCF node starts a timer upon transmitting the request; receiving, by the P-CSCF node, an indication of an error condition from the DRA or SCP prior to timeout of the timer, wherein the error condition corresponds to a network error rate exceeding an error threshold or a network performance metric failing to meet a performance threshold; and based at least on the error condition, sending, by the P-CSCF node, an instruction to the UE that instructs the UE to use an existing 5G QoS identifier 5 (5QI-5) bearer for the voice session of the UE, wherein the existing 5QI-5 bearer is a non-GBR data pipe of the 5G network.
15. The one or more computer storage devices of claim 14, wherein the 5QI-1 or 5QI-2 bearer comprises a dedicated bearer, and wherein the existing 5QI-5 bearer comprises a default bearer.
16. The one or more computer storage devices of claim 14, wherein the DRA or SCP forwards the request to a policy and charging rules function (PCRF).
17. The one or more computer storage devices of claim 14, wherein the DRA or SCP forwards the request to a policy control function (PCF).
18. The one or more computer storage devices of claim 14, wherein the voice session is a voice

over long-term evolution (VoLTE) call.

19. The one or more computer storage devices of claim 14, wherein the voice session is a voice over new radio (VoNR) call.

20. The one or more computer storage devices of claim 14, wherein the voice session is a video call.
